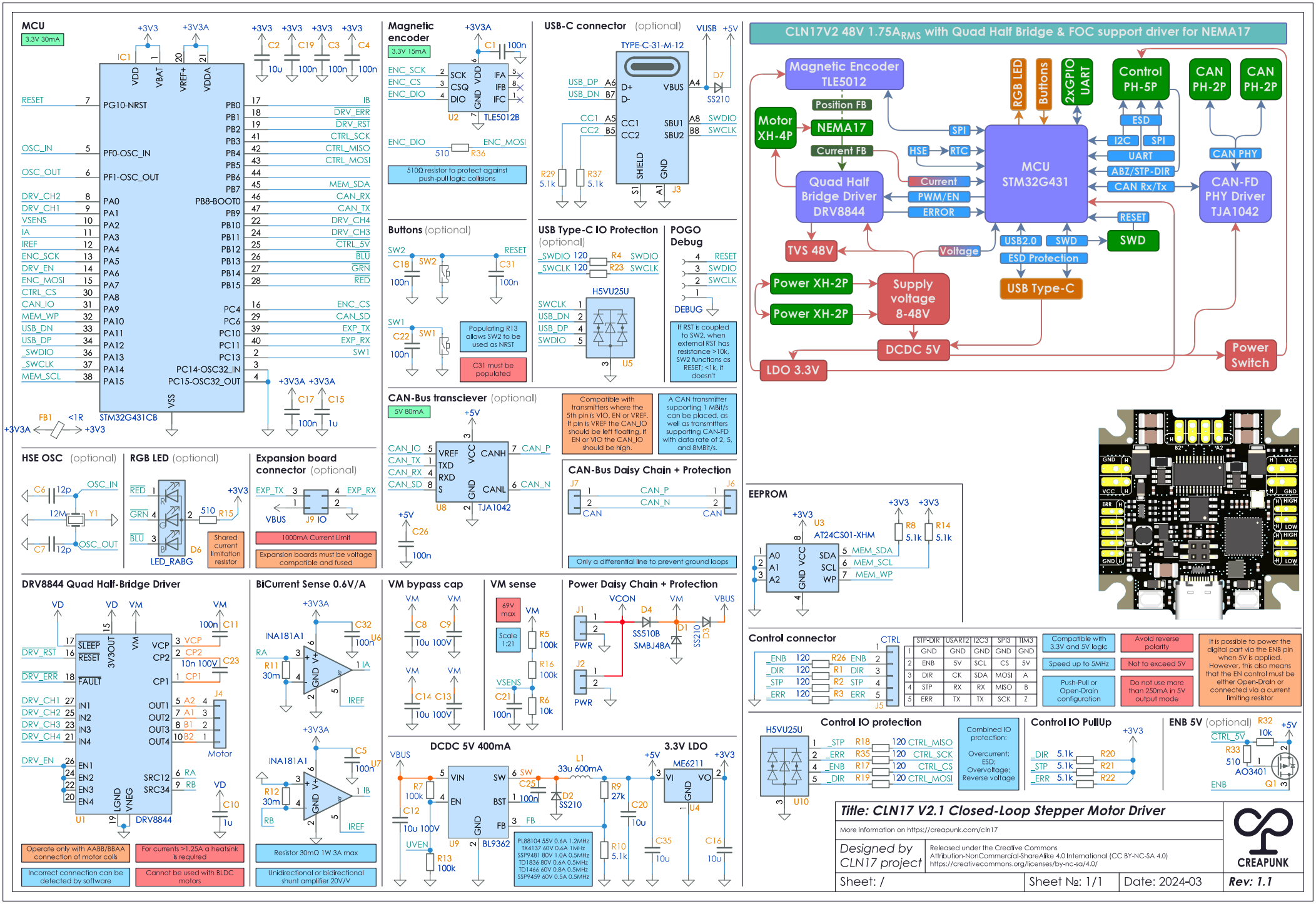

MCU
3.3V 30mA
STM32G431CB
Magnetic encoder
3.3V 15mA
TLE5012B
510Ω resistor to protect against push-pull logic collisions
USB-C connector (optional)
TYPE-C-31-M-12
Buttons (optional)
Populating R13 allows SW2 to be used as NRST
C31 must be populated
USB Type-C IO Protection (optional)
H5VU25U
POGO Debug
If RST is coupled to SW2, when external RST has resistance >10k, SW2 functions as RESET; <1k, it doesn't
CAN-Bus transciever (optional)
5V 80mA
TJA1042
Compatible with transmitters where the 5th pin is VIO, EN or VREF. If pin is VREF the CAN_IO should be left floating, if EN or VIO the CAN_IO should be high.
A CAN transmitter supporting 1 MBit/s can be placed, as well as transmitters supporting CAN-FD with data rate of 2, 5, and 8MBit/s.
CAN-Bus Daisy Chain + Protection
Only a differential line to prevent ground loops
HSE OSC (optional)
RGB LED (optional)
LED_RABG
Shared current limitation resistor
Expansion board connector (optional)
1000mA Current Limit
Expansion boards must be voltage compatible and fused
DRV8844 Quad Half-Bridge Driver
Operate only with AABB/BBAA connection of motor coils
For currents >1.25A a heatsink is required
Incorrect connection can be detected by software
Cannot be used with BLDC motors
BiCurrent Sense 0.6V/A
INA181A1
Resistor 30mΩ 1W 3A max
Unidirectional or bidirectional shunt amplifier 20V/V
VM bypass cap
VM sense
Power Daisy Chain + Protection
DCDC 5V 400mA
PL88104 55V 0.6A 1.2MHz
TX4137 60V 0.6A 1MHz
SSP9481 80V 1.0A 0.5MHz
TD1836 80V 0.6A 0.5MHz
TD1466 60V 0.8A 0.5MHz
SSP9459 60V 0.5A 0.5MHz
3.3V LDO
ME6211
CLN17V2 48V 1.75A$_{RMS}$ with Quad Half Bridge & FOC support driver for NEMA17
Magnetic Encoder TLE5012
Position FB
Motor XH-4P
NEMA17
Current FB
Quad Half Bridge Driver DRV8844
MCU STM32G431
RGB LED
Buttons
2xGPIO UART
Control PH-5P
CAN PH-2P
CAN PH-2P
ESD
I2C
SPI
UART
ABZ/STP-DIR
CAN Rx/Tx
CAN PHY
CAN-FD PHY Driver TJA1042
SPI
HSE
RTC
Current
PWM/EN
ERROR
RESET
SWD
USB2.0
SWD
ESD Protection
USB Type-C
Voltage
TVS 48V
Power XH-2P
Power XH-2P
Supply voltage 8-48V
DCDC 5V
LDO 3.3V
Power Switch
EEPROM
AT24CS01-XHM
Control connector
Compatible with 3.3V and 5V logic
Speed up to 5MHz
Push-Pull or Open-Drain configuration
Avoid reverse polarity
Not to exceed 5V
Do not use more than 250mA in 5V output mode
It is possible to power the digital part via the ENB pin when 5V is applied. However, this also means that the EN control must be either Open-Drain or connected via a current limiting resistor
Control IO protection
Combined IO protection: Overcurrent; ESD; Overvoltage; Reverse voltage
Control IO PullUp
ENB 5V (optional)
Title: CLN17 V2.1 Closed-Loop Stepper Motor Driver
More information on https://creapunk.com/cln17
Designed by CLN17 project
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CREAPUNK
Sheet: /
Sheet №: 1/1
Date: 2024-03
Rev: 1.1